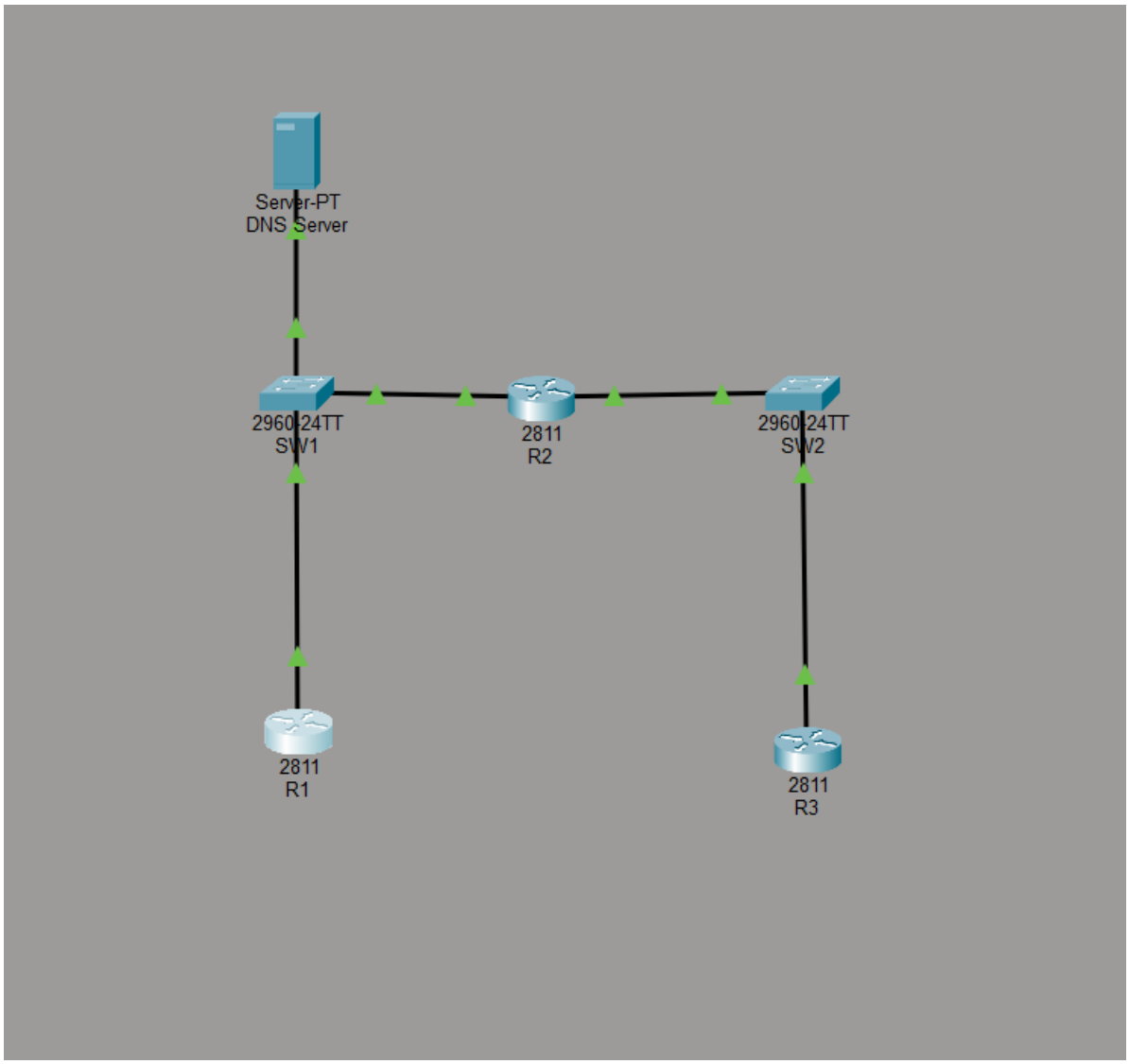
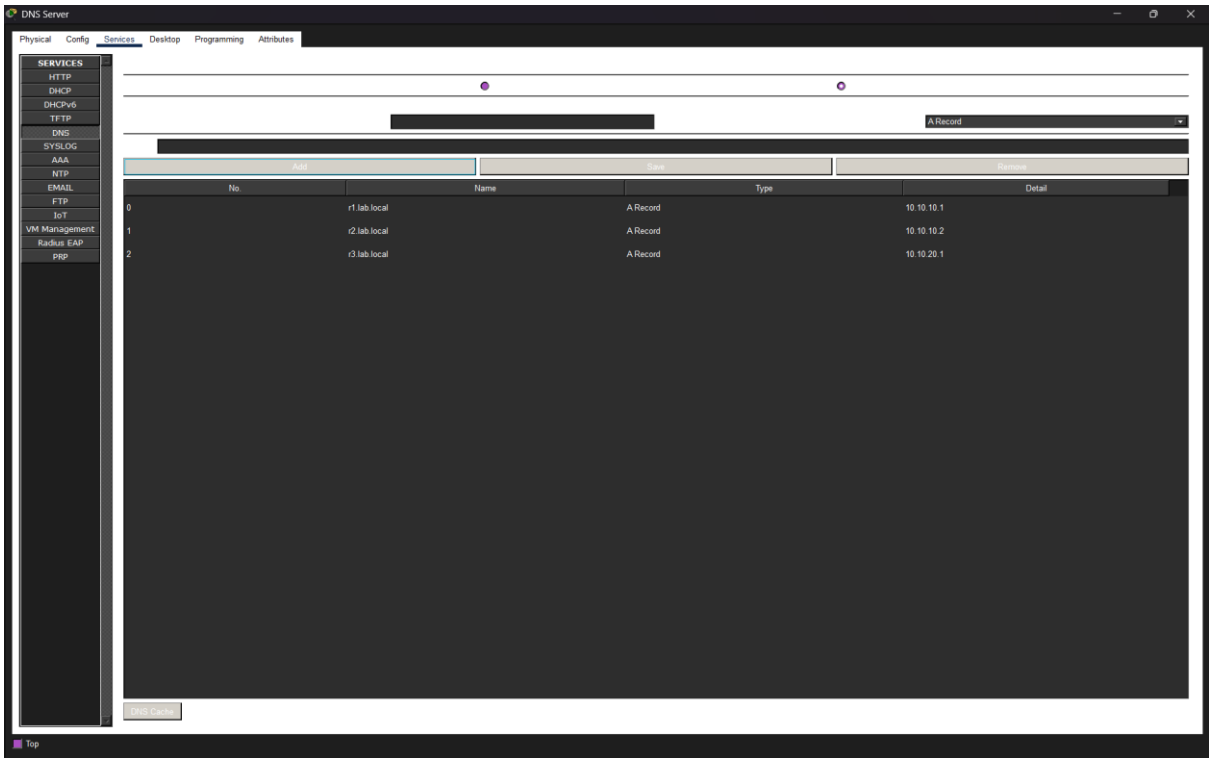
Topology


Server-PT
DNS Server
2960-24TT
SW1
2811
R2
2960-24TT
SW2
2811
R1
2811
R3

DNS Server and Records



Router DNS Client Configuration (Task 16)



Successful DNS Name Resolution (Task 17)

```
R1#ping r2.lab.local
Translating "r2.lab.local"...domain server (10.10.10.10)
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

R1#
```

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ARP Cache Verification (Tasks 18–19)

```
R1#enable
R1#clear arp-cache
R1#ping 10.10.10.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

R1#show arp
Protocol  Address          Age (min)  Hardware Addr   Type   Interface
Internet  10.10.10.1              -   000C.85E1.A801  ARPA   FastEthernet0/0
Internet  10.10.10.2              0   00E0.F920.5E01  ARPA   FastEthernet0/0
Internet  10.10.10.10             0   00E0.B0EC.8042  ARPA   FastEthernet0/0
R1#
```

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Written Answers (Questions 20–21)

An ARP entry shows the MAC address linked to an IP address on the local network, and a routing table only shows where packets should be sent, but it does not show the device's mac address.

A router uses its routing table to find where to send a packet, but it still needs the mac address of the next device, ARP helps the router find this mac address so it can send the packet over the network.